

90-DAY RAG EXPERT JOURNEY - MASTER PLAN

Start Date: January 17, 2026

Current Day: 15/90 (17% complete)

Status: Week 3 - Production Readiness

OVERALL MISSION

Become a RAG expert in 90 days while building in public. Ship production-grade systems, build audience, document everything.

PROGRESS TRACKER

Week 1: Foundation (Days 1-7)

- ☒ Day 1: RAG basics, first implementation
- ☒ Day 2: Embeddings comparison (OpenAI vs local)
- ☒ Day 3: Chunk size optimization (200 vs 500 vs 1000)
- ☒ Day 4: RAGAS evaluation framework
- ☒ Day 5: Retrieval optimization
- ☒ Day 6: Article #1 (2,857 words, Medium)
- ☒ Day 7: Week 1 review + rest

Week 2: Advanced Techniques (Days 8-14)

- ☒ Day 8: Multi-Query RAG (parallel search)
- ☒ Day 9: HyDE (hypothetical documents)
- ☒ Day 10: Self-RAG (quality gating)
- ☒ Day 11: CRAG (web search fallback)
- ☒ Day 12: Agentic RAG (iterative loops)
- ☒ Day 13: Newsletter #2
- ☒ Day 14: Article #2 (3,500 words, Medium)

Week 3: Production Readiness (Days 15-21)

- ☐ Day 15: Caching strategies (TODAY)
- ☐ Day 16: A/B testing framework
- ☐ Day 17: Monitoring & observability
- ☐ Day 18: Cost optimization

- ☐ Day 19: Error handling & graceful degradation
- ☐ Day 20: Newsletter #3
- ☐ Day 21: Week 3 review

Weeks 4-12: Advanced Topics (Days 22-90)

Plan updates weekly based on learning

TECHNICAL STACK

Models:

- Gemma 3 12B (30 RPM) - Primary
- Gemini 2.5 Flash Lite (10 RPM) - Fallback
- text-embedding-3-small (OpenAI)

Infrastructure:

- ChromaDB (vector database)
- Python 3.9
- GitHub: github.com/jugalsheth/my-first-rag

Evaluation:

- RAGAS framework
- Manual grading (1-5 scale)
- Cost tracking (\$)

Knowledge Base:

- 13 chunks (RAG papers 2024)
 - Will expand in Week 4
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CONTENT CADENCE

Daily:

- 1 LinkedIn post (8 AM ET)
- 1 Twitter post (9 AM ET)

- Code pushed to GitHub

Weekly:

- 1 Newsletter (Sunday)
- 1 Major article (Saturday/Sunday)

Platform URLs:

- Newsletter: jugals.substack.com
 - LinkedIn: linkedin.com/in/jugalsheth
 - Twitter: twitter.com/jugalsheth
 - Medium: [@jugalshethpro](https://medium.com/@jugalshethpro)
 - GitHub: github.com/jugalsheth/my-first-rag
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KEY LEARNINGS (So Far)

Technical

1. Rate limits are real constraints (plan for them)
2. Better models > complex code (Gemma vs Gemini)
3. 3-tier routing beats binary (CRAG insight)
4. Small KB needs different thresholds (13 chunks)
5. Fallback scoring must be conservative
6. 30-40% queries are repeats (cache opportunity)

Content

1. 8-word hooks (Lara's framework)
2. SLAY: Story → Lesson → Actionable → You
3. PAS: Problem → Agitate → Solution
4. Engage first 30 mins after posting
5. Real data > theory
6. Honest failures build trust

Audience

1. Consistency > perfection (14 days straight)

2. Newsletter = owned audience
 3. LinkedIn > Twitter for tech content
 4. Show work, not just results
 5. Reply to every comment
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BUDGET TRACKING

Monthly Budget: \$20 **Current Spend:** ~\$5 (Days 1-14) **Projected:** \$15/month at current rate

Cost Breakdown:

- API calls (Gemini/Gemma): \$3
 - Embeddings (OpenAI): \$1
 - Web search (Tavily): \$1
 - Total: ~\$5/month 
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17 WEEK 3 DETAILED PLAN

Day 15: Caching Strategies

Goal: Make RAG 10x faster for repeated queries

Build:

- Tier 1: Exact match cache (hash-based)
- Tier 2: Semantic similarity cache (embedding)
- Tier 3: Full RAG (fallback)

Test:

- 20 queries (10 unique, 10 repeats)
- Measure: latency, cost, hit rate

Expected:

- 40-60% hit rate
- 50% latency reduction
- 50% cost savings

Posts:

- LinkedIn: "Made RAG 10x faster by caching"
- Twitter: Thread on 3-tier caching

Day 16: A/B Testing Framework

Goal: Compare techniques scientifically

Build:

- Test harness for comparing RAG variants
- Metrics: accuracy, cost, latency
- Statistical significance tests

Test:

- Multi-Query vs Standard
- HyDE vs Standard
- Cached vs Uncached

Posts:

- "Which RAG technique wins? Data says..."

Day 17: Monitoring & Observability

Goal: Track production metrics

Build:

- Logging system (queries, errors, latency)
- Metrics dashboard
- Alert system for failures

Test:

- Simulate production load
- Track error rates
- Identify bottlenecks

Day 18: Cost Optimization

Goal: Reduce production costs

Build:

- Cost tracking per query
- Batch processing for embeddings
- Smart caching policies

Test:

- Measure savings from optimizations
- ROI analysis

Day 19: Error Handling

Goal: Graceful degradation

Build:

- Retry logic with backoff
- Fallback responses
- Circuit breakers

Test:

- Simulate API failures
- Rate limit scenarios
- Network issues

Day 20: Newsletter #3

Topic: "Production RAG: Performance & Reliability"

Content:

- Week 3 synthesis
- Caching strategies
- Monitoring approach
- Cost optimization
- Error handling

Day 21: Week 3 Review

Rest day + planning

- Review Week 3 learnings
 - Plan Week 4 topics
 - Analyze metrics (followers, engagement)
 - Adjust strategy
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SUCCESS METRICS

Technical (by Day 90)

- ☐ 10+ RAG techniques implemented
- ☐ Production-ready system
- ☐ <100ms p95 latency
- ☐ 99.9% uptime
- ☐ <\$50/month cost

Content (by Day 90)

- ☐ 90 LinkedIn posts
- ☐ 90 Twitter posts
- ☐ 12 newsletters
- ☐ 4-6 major articles
- ☐ GitHub repo with 50+ stars

Audience (by Day 90)

- ☐ 1,000+ LinkedIn followers
 - ☐ 500+ Twitter followers
 - ☐ 100+ newsletter subscribers
 - ☐ 5,000+ article views
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TOOLS & FRAMEWORKS

Writing:

- SLAY Framework (LinkedIn)
- PAS Framework (Twitter)

- 8-word rule (hooks)
- 30-minute engagement rule

Development:

- Cursor + Claude (AI pair programming)
- GitHub (version control)
- ChromaDB (vector DB)
- RAGAS (evaluation)

Content:

- Substack (newsletter)
 - Medium (articles)
 - LinkedIn (primary)
 - Twitter (secondary)
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REFERENCE MATERIALS

Key Papers:

- RAG Survey (2024)
- RAGAS paper
- HyDE paper
- Self-RAG paper

Resources:

- ByteByteGo (system design)
 - Lara Acosta (content frameworks)
 - NotebookLM (paper comprehension)
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IMPORTANT REMINDERS

1. **Never skip a day** (consistency > perfection)
2. **Post before coding** (audience building)

3. **Real data always** (no theory-only posts)
 4. **Engage 30 mins** after posting
 5. **Document failures** honestly
 6. **Update GitHub daily**
 7. **Track costs** weekly
 8. **Review progress** weekly
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MILESTONES

- ☒ Day 7: Week 1 complete
 - ☒ Day 14: Week 2 complete
 - ☐ Day 21: Week 3 complete
 - ☐ Day 30: Month 1 complete
 - ☐ Day 60: Month 2 complete
 - ☐ Day 90: Journey complete
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LINKS QUICK REFERENCE

Content:

- Newsletter: <https://jugals.substack.com>
- Article 1: [Week 1 Medium URL]
- Article 2: <https://medium.com/@jugalshethpro/advanced-rag-techniques-what-actually-works-in-production-2de1b0ff2fe8>

Code:

- Main repo: <https://github.com/jugalsheth/my-first-rag>
- Week 1 code: /Day1-7 folders
- Week 2 code: /Day8-14 folders

Social:

- LinkedIn: <https://linkedin.com/in/jugalsheth>
 - Twitter: <https://twitter.com/jugalsheth>
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UPDATE LOG

- 2026-01-17: Journey started (Day 1)
 - 2026-01-23: Week 1 complete
 - 2026-01-27: Newsletter #1 published
 - 2026-01-28: Week 2 complete, Article #2 published
 - 2026-01-29: Week 3 started (Day 15)
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Last Updated: 2026-01-29 (Day 15) **Next Update:** Weekly or as needed **Status:** On track, 15/90 days complete (17%)